

Выпуклые множества:

Каратеодори, Радон, Хелли

Homework

1 сентября 2026 года

1. (Jung 1901) Prove that a set of diameter 1 in $\mathbb{R}^n$ is contained in a ball of radius $\sqrt{n/(2n+2)}$, and that the regular simplex shows this is optimal. (*Hint: apply Helly to the balls centred at the points of the set.*)
2. (Steinitz 1913) Let x be an interior point of $\text{conv } S$, $S \subseteq \mathbb{R}^n$. Prove that x is an interior point of $\text{conv } S'$ for some $S' \subseteq S$ with $|S'| \leq 2n$, and that $2n$ is optimal.
3. (Bárány 1982) Prove the colorful Carathéodory theorem: if $x \in \text{conv } S_i$ for $S_1, \dots, S_{n+1} \subseteq \mathbb{R}^n$, then one can pick $x_i \in S_i$ with $x \in \text{conv}\{x_1, \dots, x_{n+1}\}$. (*Hint: take the “rainbow simplex closest to x .”*)
4. (Tverberg 1966) Prove Tverberg’s theorem on the line: any $2r-1$ points of $\mathbb{R}$ split into r groups whose convex hulls have a common point. State the general case and check that $r=2$ is Radon’s theorem.
5. (Rado 1946) Deduce the centerpoint theorem from Helly: for any N points in $\mathbb{R}^n$ there is a point c such that every closed halfspace containing c contains at least $N/(n+1)$ of them.
6. (Капачёв 3.52) Let Q_1, Q_2 be quadratic forms on $\mathbb{R}^n$, $n > 2$, and let $Q = (Q_1, Q_2): \mathbb{R}^n \rightarrow \mathbb{R}^2$. Prove that the image of the unit sphere under Q is a convex set.
7. (Erdős–Szekeres 1935) Prove that any five points in general position in the plane contain four in convex position. State the general «happy ending» problem and explain what is known about the number $f(n)$ of points forcing a convex n -gon.

8. (Hadwiger–Debrunner) Finitely many segments in the plane are parallel to the y -axis and for any three of them there is a line meeting all three. Prove that some line meets all of them. (*Hint: “the line $y = ax + b$ meets the segment is a pair of linear inequalities in (a, b) .”*)
9. (ФКН, осень 2024) Let $K_1, \dots, K_n$ be convex sets in $\mathbb{R}^d$ such that the unit ball B centred at the origin meets the intersection of any k of them. Prove that there is a point $x \in \mathbb{R}^d$ with $\text{dist}(x, K_j) \leq 1/\sqrt{k}$ for every j . (*Hint: replace K_j by $K_j \cap B$ and rewrite the condition through support functions; averaging over a random set of k indices gives exactly $1/\sqrt{k}$.*)
10. (VJIMC 2017, cat. I) Let P be a convex polyhedron; to each vertex one assigns a nonnegative number so that the numbers sum to 1, and to each edge the product of the numbers at its endpoints. Prove that the sum over the edges is at most $3/8$. (*Hint: this is the Motzkin–Straus theorem on the maximum of the quadratic form of a graph over the simplex.*)
11. (Kirchberger 1903) Finite sets $A, B \subseteq \mathbb{R}^n$ can be strictly separated by a hyperplane if and only if for every $C \subseteq A \cup B$ with $|C| \leq n + 2$ the sets $C \cap A$ and $C \cap B$ can be strictly separated.
12. (Doignon 1973) Prove the lattice Helly theorem for $n = 1$ and $n = 2$: if any 2^n of the convex sets $K_1, \dots, K_m \subseteq \mathbb{R}^n$ have a common point of $\mathbb{Z}^n$, then all of them do; show that 2^n is optimal for $n = 2$.
13. (IMC 2025 shortlist) Let $A_1 A_2 \dots A_{2025}$ be a convex polygon and let B be a set of points such that for all $i < j < k$ at least one point of B lies strictly inside the triangle $A_i A_j A_k$. Find the minimal possible number of points of B .
14. (Katchalski–Liu 1979, one-dimensional case) Prove the fractional Helly theorem on the line: if at least $\alpha \binom{m}{2}$ of the pairs of m intervals intersect, then some $\alpha m/2$ of the intervals have a common point. State the theorem in $\mathbb{R}^n$ and check that your proof does not generalise verbatim.